SpiceMagic -- Evidence Grading Layer

Attaching an evidence tier to every one of 228,750 spice-disease associations, then re-scoring the database weighted by quality of evidence.

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1. The problem

Our current database records claims like (turmeric, cancer, POSITIVE) and, in aggregation, counts every one of those rows equally. But the rows are not equal. One may come from "curcumin induced apoptosis in HCT-116 colon-cancer cells at 25 uM" -- a cell line in a dish. Another from "a randomized double-blind trial of 120 patients found no significant difference in tumour progression." Treating these as one vote each is misleading.

Because in-vitro studies are cheap and vastly outnumber human trials, the aggregate always tilts positive -- this is a large part of the 56% positive share in our results. Crucially, this is not only reporting bias: it is evidence-tier composition, and unlike publication bias, that part is measurable and correctable. Turmeric is the textbook case -- an enormous in-vitro literature, but human trials repeatedly limited by very poor oral bioavailability. Our database currently cannot express that distinction at all.

2. The goal

Attach an evidence tier to every association, then recompute our headline findings weighted by tier -- and always show the old (unweighted) and new (tier-weighted) numbers side by side. The question we can finally answer: when we look only at human evidence, does the 56% positive picture survive, or collapse toward neutral?

3. The evidence-tier system (six levels)

We grade each study by where it sits on the standard evidence pyramid, from strongest human evidence (tier 1) to weakest / non-primary evidence (tier 6). Six coarse, well-separated classes classify far more reliably than a dozen fine-grained ones.

Tier	Class	What it means	Example sentence
1	Synthesis	Meta-analysis or systematic review of human trials	"A meta-analysis of 12 randomized trials of curcumin..."
2	RCT	Randomized controlled trial in humans	"randomized, double-blind, placebo-controlled trial of 120 patients"
3	Human (non-random)	Cohort, case-control, cross-sectional, observational, or case report	"a survey of 340 patients"; "a case report of anaphylaxis after..."
4	Animal	In-vivo animal study (mice, rats, murine, zebrafish)	"administered orally to Wistar rats for 8 weeks"
5	In-vitro	Cell line / biochemical assay (no organism)	"induced apoptosis in HCT-116 cells at 25 uM"
6	Review / in-silico	Narrative review, opinion, or computational only	"this review summarizes..."; "molecular docking of piperine"

Why the order: a result seen in a dish (tier 5) or a mouse (tier 4) is far weaker evidence of human benefit than a randomized trial (tier 2) or a meta-analysis (tier 1). Case reports are also flagged separately, since they are mostly adverse-event (harm) reports rather than effectiveness studies.

4. How we assign the tiers (cheaply)

- **Free metadata signal:** PubMed already tags Meta-Analysis, Systematic Review, Randomized Controlled Trial, Clinical Trial and Observational Study. One metadata pass gives us tiers 1-3 for free.

- **Text rules:** Metadata cannot tell animal from in-vitro, so we add high-precision keyword rules over the abstract - "IC50", "uM", cell-line names for tier 5; "mice", "Wistar rats", "in vivo" for tier 4; "patients", "participants" for tier 3.

- **Gold set:** ~1,000 papers hand-labelled to validate the rules (measure their precision) and to cover the cases rules miss.

- **Classifier:** The same PubMedBERT architecture as the capstone, re-trained for six-class tier prediction, fills in every remaining paper.

So far the metadata + text rules already label 83.5% of our 49,499 papers with high precision; the classifier handles the rest.

5. Preliminary finding (rules only, to be confirmed)

Even before the classifier and hand-validation, joining the tier labels to our sentiment database (88% of associations covered) already shows the effect clearly:

Evidence tier	Associations	% positive
1 Synthesis	5,572	33.0%
2 RCT	10,631	42.0%
3 Human (non-random)	34,102	18.6%
4 Animal	73,056	75.2%
5 In-vitro	41,556	58.1%
6 Review / in-silico	37,342	59.5%

Read the pattern: animal studies report benefit 75% of the time and almost never report a null result, but human trials are positive only 42% of the time, and human observational studies just 19%. As the evidence gets closer to real patients, the benefit signal shrinks -- textbook translational optimism.

Headline: the raw 56.3% positive falls to 25.2% when restricted to human evidence (tiers 1-3). Preclinical and review work make up 75% of the database.

(Preliminary, from rules alone. Final numbers follow the hand-validated classifier; direction is not expected to change.)

6. What the finished project delivers

- The sentiment split recomputed on human evidence only - does 56% positive hold?
- The beneficial-spice ranking recomputed on tiers 1-3 - which spices are famous only from cell-culture work, and drop out?
- A tier-composition bar per top spice - which spices actually have human trials behind them (the single strongest figure).
- A controversy atlas - pairs where cheap in-vitro optimism disagrees with human evidence (the curcumin / turmeric case).
- New evidence-tier filters and an "what is the evidence really like" view in the web app.

7. Status

Metadata harvested for all 49,499 papers; six-tier rule labeller built and covering 83.5% of papers; the 1,000-paper gold annotation set is prepared and ready to label. Next: hand-label the gold set, train the tier classifier, run full inference, and produce the tier-corrected re-analysis.